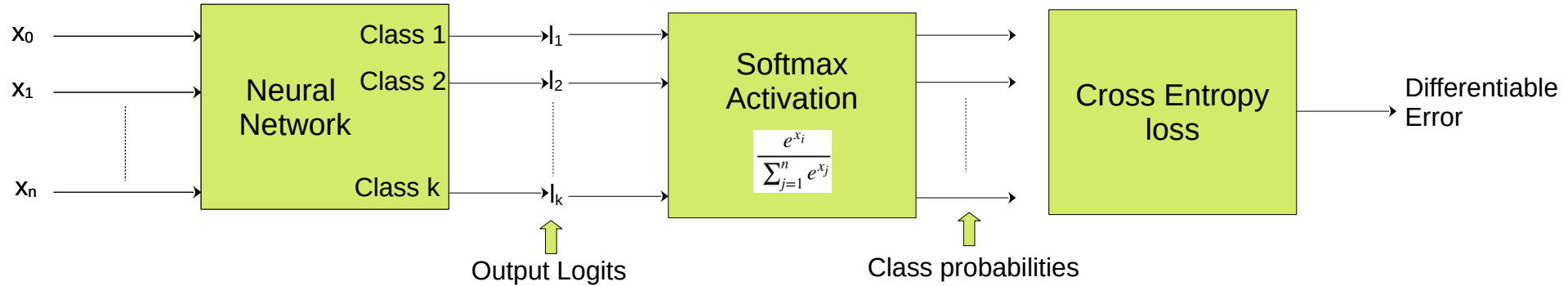


Softmax and Cross Entropy in Classification

Goals:

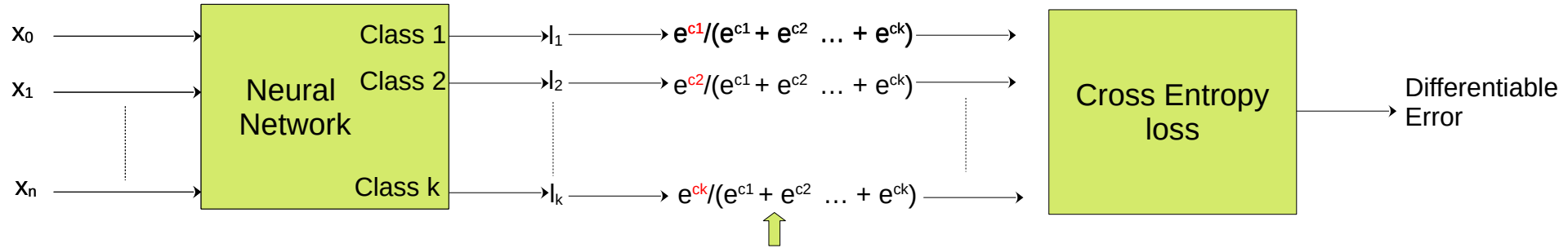
- Explain what softmax does and why we use it
- Define cross-entropy and its gradient
- Implement the pair correctly for multi-class classification

Where They Fit in a Classifier



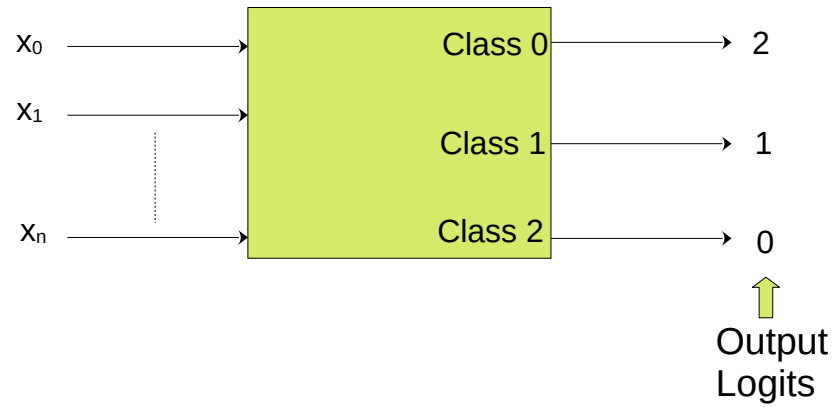
- **Logits**: arbitrary real scores from model (unnormalized)
- Largest logit is the winner
- **Softmax** outputs class 'probabilities'
- **Cross Entropy** how wrong we are.

Softmax Activation

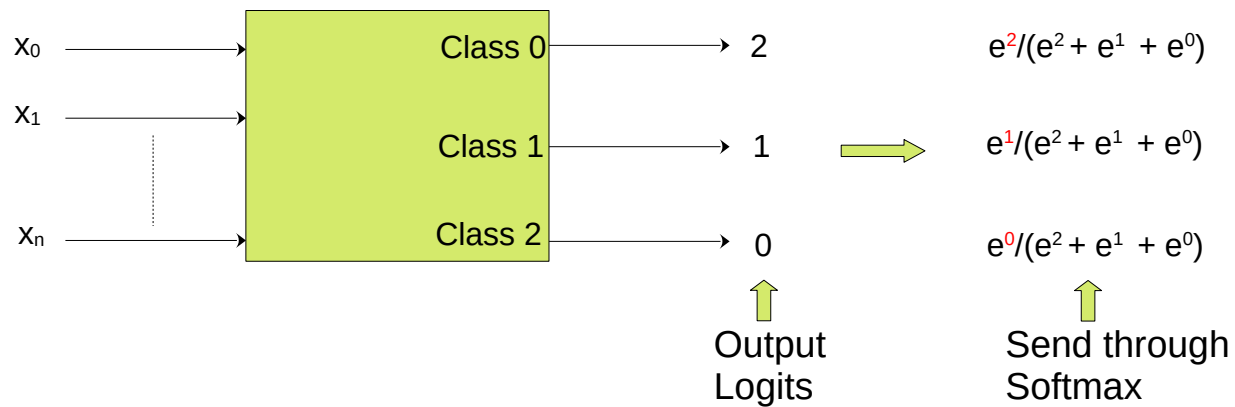


- Turns raw scores (“logits”) into a ‘probability’ like distribution (sum to 1)
- **Differentiable**
- Largest softmaxed logit is still largest
- In fact softmax pushes largest higher and others lower

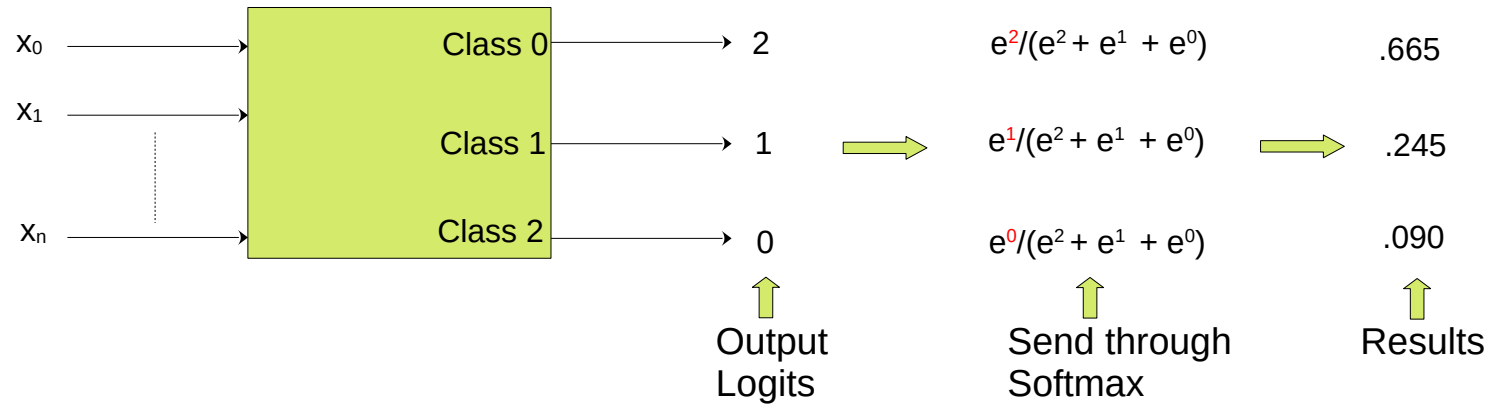
Softmax Activation Example



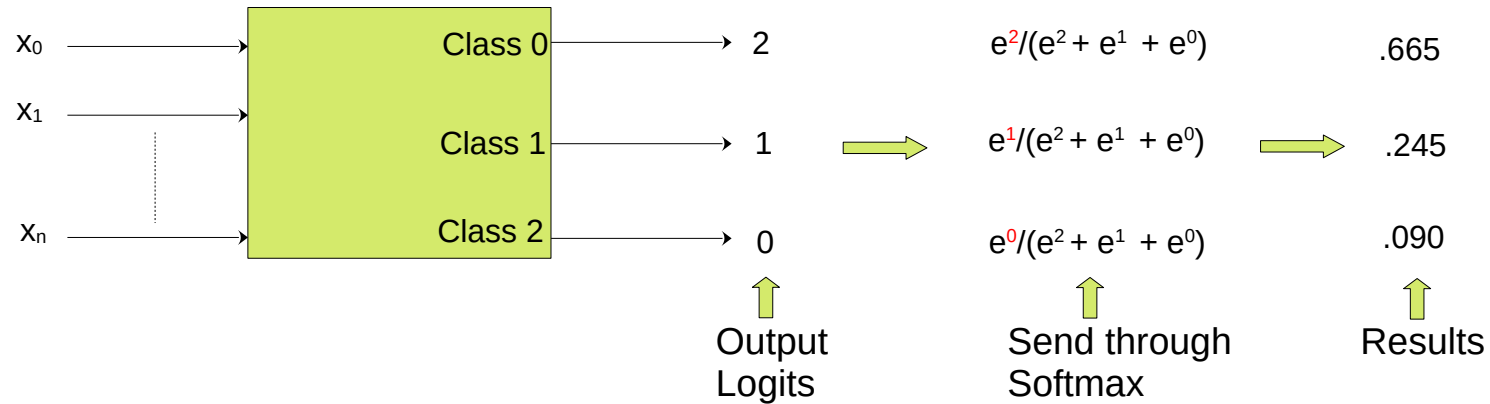
Softmax Activation Example



Softmax Activation Example



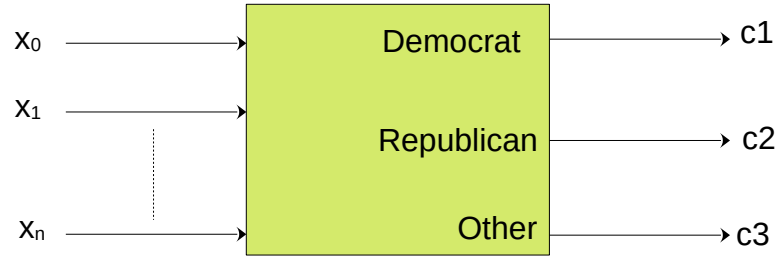
Softmax Activation Example



- $.665 + .245 + .090 = 1$
- Class 0 is still selected
- Class 0 probability is much larger than the original gap between Class 0 and Class 1 suggests

Softmax Classification Example

use argmax to select predicted class



A NN with a final layer softmax activation function.
Three training rows are fed in with the following results

c1	c2	c3	targets	correct?
0.3	0.3	0.4	0 0 1 (democrat)	yes
0.3	0.4	0.3	0 1 0 (republican)	yes
0.1	0.2	0.7	1 0 0 (other)	no

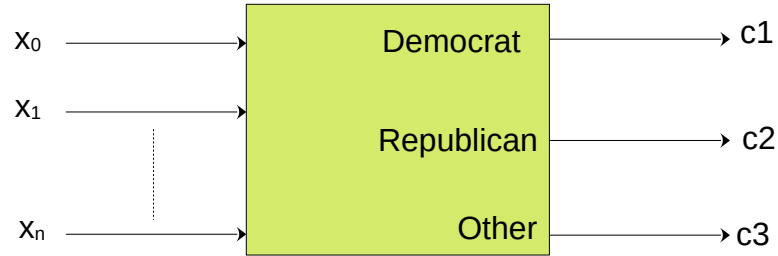
It has a classification accuracy of 2/3 or 66%

It has a classification error of 1/3 or 33%

It barely gets the first 2 right and is way off on the third.

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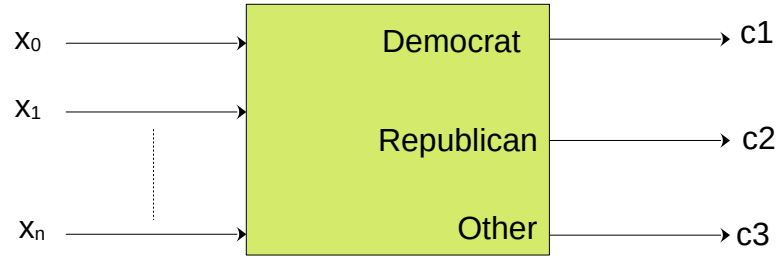
A different NN with a final layer softmax activation function.
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It nails the first 2 and barely misses the third.

The second network is clearly better but the metrics do not reflect that
Also, argmax is not differentiable so cannot backprop through it

Cross Entropy Classification Error

Why it is better

- Also called the **negative log** likelihood
- Takes the negative log of the softmax score for ONLY the correct class. All other entries are ignored.
- So applying cross entropy error for the previous 2 examples;

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Row1 error= $-(\ln(0.3)*0) + (\ln(0.3)*0) + (\ln(0.4)*1) = -\ln(0.4)$

Row2 error= $-\ln(.4)$

Row3 error= $-\ln(.1)$

Total error= $-(\ln(0.4) + \ln(0.4) + \ln(0.1)) / 3 = 1.38$

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The error now reflects that the second network is much better than the first

Pytorch Cross Entropy Example

```
criterion = nn.CrossEntropyLoss()
```

```
def train(model, train_loader, test_loader, optimizer, epochs=10):  
    for ep in range(1, epochs+1):  
        model.train()  
        run_loss, n = 0.0, 0  
        for xb, yb in train_loader:  
            xb, yb = xb.to(device), yb.to(device)  
            optimizer.zero_grad(set_to_none=True)  
            logits = model(xb)  
            loss = criterion(logits, yb)  
            loss.backward()
```

Note: pytorch cross entropy will automatically apply softmax before it calculates the cross entropy (SO DONT ADD A SOFTMAX LAYER)

When not to use Cross Entropy

- Multi-label problems (can have more than one label)
- Instead feed logits into BCEWithLogitsLoss

Summary

- For single label classification use cross entropy loss (pytorch implementation automatically applies softmax to the input logits)
- For multilabel classification use BCEWithLogitsLoss